Getting started with spreadsheets.

1. Basic Computation
   (a) To add 3 + 4, enter `=3+4`
   (b) To subtract 100-76, enter `=100-76`
   (c) To multiply 4 times 18, enter `=4*18`
   (d) To divide 0.05 by 12, enter `=0.05/12`
   (e) To calculate , enter `=5^25`
2. Compute an 18% tip on a $35.75 bill.
   (a) What is 18% as a decimal? 0.18
   (b) What should we enter into the spreadsheet? (35.75)(0.18) = $6.44
3. Make a tip calculator using **cell references**.
   (a) Open a new spreadsheet and label it `TIP`.
   (b) In cells A1, B1, C1, D1, type column headers:

Bill Amount	Tip Percent	Tip Paid	Total

   (c) In cells A2, B2, C2, D2, type:

$= 35.75$	$= 0.18$	$= A2 * B2$	$= A2 + C2$

   (d) What calculation did we do?

   The tip and total paid for $35.75 bill and 18% tip.

   (e) Use the calculator to determine the Total if the Bill Amount is $76.80 and the Tip Percent is 20%.

   Replace A2 with 76.80 and B2 with 0.2

4. We now demonstrate the **Fill Down** feature.
   (a) Enter 10 into call `A2` and enter `A2`+10 into cell `A3`
   (b) Select `A3` and drag down until you get to 110.
   (c) Drag down cells `B2` through `D2`.
   (d) How much is the tip on a $110 meal? $22.00

   How much is the final bill? $120.00

   How large does the Bill Amount need to be for the tip to be more than $15? $80.00

5. Make a Simple Interest Calculator

Suppose \$1000 is invested in a savings account that earns 3.2% interest on the \$1000 each year.

(a) Open a new spreadsheet called `Interest`.

(b) In cells A1, B1, C1, D1, E1 type column headers:

Interest Rate	Principal	Year	Interest	Total

(c) Determine what should be in cells A2, B2, C2, D2, E2:

=0.032	= \$1000	Start	none	

(d) Determine what should be in cells A3, B3, C3, D3, E3:

= 0.032	= \$1000	= 1	=A3*B3	=B3+D3

(e) Determine what should be in cells A4, B4, C4, D4, E4:

= 0.032	= \$1000	=C3+1	=A4*B4	= E3+D4

(f) Use the pull down feature to determine the Total in the savings account in 10 years. Write down your steps.

- highlight row 4. Pull down to row 12.
- read off E12
- Ans: \$1320.00

(g) How long will it take for the savings account to double in value? Write down your steps.

- Pull down many rows.
- In row 33, the total is \$1992. In row 34, the total is \$2024.00
- So it takes 32 years.

(h) How can you calculate the Total in the savings account in 50 years **without** using the pull down feature? Then check your calculation using the pull down feature.

Each year, the account gains \$ 32.00. So, in 50 years, the account gains (50)(\$ 32) = \$1600. So the total is \$1000 + \$1600 = \$2600. (Look in row 52 ✓)

(i) If P is the principal, r is the interest rate, t is the number of years, and A is the accumulated amount of money in the savings account, write a formula for A and check your answer.

$$A = \underset{\text{principal}}{P} + \underbrace{P \cdot r}_{\text{interest each year}} \cdot \underset{\text{number of years}}{t} = P(1 + rt) = \underset{\text{principal}}{P} + \underset{\text{interest}}{I}$$

6. How To Export the **entire workbook** as a **PDF**.
   (a) Save your work!
   (b) In upper left menu, click `File` to produce a dropdown menu.
   (c) In dropdown menu, select `Download`
   (d) In right-side menu, select `PDF`

7. Quick Summary of Finance Terms
   (a) **principal** - P - starting amount of money, present value

   (b) **interest rate** - r - a percent
   - the percent of principal gained in a given time period

   (c) **interest** - I - the amount of money added each time period

   (d) **accumulated amount** or **end amount** or **principal plus interest** or **future value**

   $A = P + I$ = the starting amount and the money added.

   (e) **Formula for Simple Interest over Time**

   $$A = P(1 + rt) = P + Prt$$

   (f) Comment on time t.

   In this example, interest was added each year. But this can vary, say each month, or each quarter.